

RingIn

Master Audit & Fix Report

21 audit rounds · 214 fixes shipped · 14 APK versions · 2026-05-18

214

TOTAL FIXES

21

AUDIT ROUNDS

18

APK BUILDS

13+

FILES / 12K+ LINES

RingIn · Webstreax Technologies Pvt Ltd
Latest: APK v3.42 + Bundle 2026.0518.1701

Executive summary

Started with one master audit (5 parallel agents → 152 findings across HomeScreen, Moments, Messages, Profile, Settings, Calls, Wallet, App.js, OTA, security). Shipped fixes in batches, then re-audited with a fresh angle each round. Every round still surfaced real user-impact bugs — the 12,000+ line codebase had genuine surface area.

Per-round trend

R7 : 9 · R8 : 8 · R9 : 10 · R10 : 8 · R11 : 11 · R12 : 6 · R13 : 8 · R14 : 10 · R15 : 8 · R16 : 9 · R17 : 8 · R18 : 11 · R19 : 10 · R20 : 8 · R21 : 7

Initial batches (R1-6): 83 fixes from the master audit. Average per fresh-angle round (R7+): ~8.6 fixes.

Process discipline

Every batch followed the same loop:

1. **Audit agent** — independent agent reads the actual code, finds bugs with file:line citations
2. **Fix agent** — receives explicit instructions per item, edits surgically, NO JSX, var pattern, hook order preserved
3. **Verification agent** — independent agent checks each fix for scope leaks, hook order violations, regression of prior fixes
4. **Build check** — full CRA build must compile clean before ship
5. **Ship** — strip-build-for-apk lean APK + OTA bundle to GitHub raw + tracker PDF update + push to main

Why fresh angles each round

After round 6 ship, the audit agent kept reporting "Clean" against the same questions. So each subsequent round was given an **orthogonal angle**: different file-set focus (R9), different bug class (R10–R17). Each new angle found genuine bugs the previous rounds had missed. Suggests that any "no findings" result is more a sign of stale angles than a clean codebase.

Round-by-round breakdown

P0

3 fixes

Crash-on-tap scope leaks

- **UserProfileView Edit Post + Mute** referenced HomeScreen-scoped vars → ReferenceError on tap
- **ProfileScreen onViewUser / onSwitchTab** used props.X but component destructured props — undefined

B1

18 fixes

ProfileScreen — bio merge + dummy buttons + small honest fixes

- Bio JSON deep-merge in Save All + Expert App + Account Settings prefill (was wiping cover_url, location, notif_prefs, sound_prefs every save — silent data loss)
- 8 dummy buttons converted: Edit Post, Mute, Photo, GIF, Set Reminder, Connect Voice, Saved Posts click, Help & Support
- InstallPrompt accept-not-dismiss fix, login log Array.isArray guard, Volume slider stale closure, Test Sound variant, Expert App validation, SavedPosts safety timer

B2

12 fixes

Messages + chat — realtime sync + block unification

- Realtime UPDATE / DELETE / message_reactions handlers on chat channel
- Inbox sender_id second handler + StrictMode-safe activeRef refactor
- ChatBox restrict/mute toggles broadcast events; inbox listens (storage event doesn't fire same-tab)
- blocks.js unified with legacy ringin_blocked, conversation_id 'e' regex, getPublicUrl guard, compression-fail fallback, last-seen reset, scroll-up guard

B3

14 fixes

Memory leaks + race fixes + poll early-outs

- compVideo blob revoke, carouselIdx 400ms debounce, IntersectionObserver per-render leak + infinite loop, loadMoreFeed early return
- Back handler dep array completion (postMenu, showLikers, selectedUser, compAudienceMenu)
- Moments willChange reset, muted-users filter applied, sendMomentToUser convId, auto-advance refs
- CallScreen poll early-out, App pollOnce visibilitychange, lastSeen listener cleanup, sendBeacon("") removed

B4

10 fixes

Coin sync + Wallet atomicity + StrictMode

- coinBalance per-user STORAGE_KEY (was global → multi-user leak), .maybeSingle (new users), cancelled flag, unique channel UUID
- WalletScreen near-atomic purchase (re-fetch before write), payTimerRef unmount cleanup
- CallScreen setSharedCoinBalance moved out of updater (StrictMode safe), 0-balance guard + coins-prop sync
- MessagesScreen + SearchScreen + HomeScreen coin:50 stubs removed

B5**10 fixes****Mock data → real queries + N+1 fanout**

- N+1 notification fanout → 1 batched .in() query + 1 batched insert
- Real call count from call_invites, member-since from session, real friends from follows, Skills from profileInfo.skills
- SearchScreen mock IDs namespaced 'mock_' to avoid UUID collision
- Hashtags lowercased, submitPost double-submit guard, editPostProfData added to back-handler dep array

B6**15 fixes****Polish — alerts→toasts, validation, hygiene**

- 3 alerts in App.js → toasts, RINGIN_BUILD debug-gated
- WalletScreen Luhn check, Amex 4-digit CVV, UPI regex, real ₹ conversion
- MessagesScreen + HomeScreen timeAgo no 'Z' append, msgMenu position clamp
- Dead NOTIFS array removed, getRecommendedExperts import removed, UpdatePrompt honest fallback notes, iOS detection Mac fix, debug=0 off-switch

R7**9 fixes****Post-shipout full audit**

- **saveEditProfile bio deep-merge** — Edit Profile path missed in B1, was wiping bio every save
- **[mshare] render** in chat + inbox (was raw text)
- HomeScreen image+video upload null-guard + .catch, App.js SIGNED_OUT cache cleanup, Save All timezone, lastSeen visibility ping
- 11 alerts → toasts in HomeScreen, MessagesScreen PTR scroller, MessagesScreen revoke order

R8**8 fixes****Strict filter — only real impact**

- Save All ack-before-write (was lying on network errors)
- saveEditProfile + Wallet processPayment .catch (stuck buttons on network reject)
- Wallet "You are buying" header missing price (empty div!)
- UserProfileView share guard, hardware-back-during-call hangup (ringin:back-call event), MessagesScreen alert→toast, Moments removed content: field

R9**10 fixes****Orthogonal — less-touched files + user flows**

- **CRITICAL:** SearchScreen inline Call button used legacy broken path — bypassed call_invites + FCM push (callee never rang)
- useCloseFriends mount sync, IncomingCallModal reject dedupe + await
- LiveWorkshopScreen scrollIntoView block:nearest, App.js _hscrollCache WeakMap
- blocks.js notify(), HomeScreen audience retry cap, MomentComposer revoke deferred, AnonymousConnect skip override, momentUsers notify error paths, IncomingCallModal token stash

R10**8 fixes****Every Supabase query + every ringin:* event**

- Download My Data exporting empty messages (was querying nonexistent content column)
- SavedPosts couldn't actually be opened (dispatched event had ZERO consumers — wired 3-hop chain)
- 4 silent-rollback gaps: Clear notifications, Delete moment, Delete post (Profile), Block user
- 2 missing .catch on chat send + reaction DELETE

R11**11 fixes****Rollback completeness + date/TZ + number parsing**

- Incoming call invites silently dropped if clock skew >60s (widened to 5min + handle clock-behind-server)
- 4 timeAgo functions rendered "-3m ago" on negative diffs (added Just now early return)
- ProfileScreen timeAgoProf Z-append bug R7 missed elsewhere
- Mark-all-notifications-read no rollback, 5 missing .catch (toggleLike, toggleLikeU, submitComment, saveEditPostProf, reaction INSERT), activity-log TZ consistency, coinBalance negative clamp, useFollow alert→toast + rollback toasts

R12**6 fixes****Realtime channels + permissions + concurrent actions**

- Moment file upload no MIME/size validation (OOM on huge DSLR raws)
- 6 raw clipboard sites in HomeScreen + Moments (replaced with helper + fallback)
- startOutgoingCall didn't guard incomingCallRef (could stack 2 calls)
- toggleLike/toggleLikeU rapid-tap created parallel RPC flicker (in-flight guard), notif channel name unique-per-mount

R13**8 fixes****iOS WebView + CSS z-index/safe-area + prop contracts**

- **CRITICAL:** Incoming call modal was BELOW Moments viewer — user mid-moment couldn't answer/decline a real call
- Wallet card inputs now show numeric keyboard on mobile (was full QWERTY for 16-digit card numbers)
- InstallPrompt + notifications panel respect iPhone X+ safe-area-insets
- Moments clickGuard 400ms→80ms, chat input enterKeyHint='send', avatar drag preventDefault, IncomingCallModal img onError fallback

R14**10 fixes****First-paint + form inputs + image URLs**

- 4 stuck-spinner spots (FollowersList, doSearch, AnonymousConnect, FollowersListModal — all had .then with no .catch)
- 9 Enter handlers now respect IME composition (CJK users no longer send half-composed messages)
- ~10 image sites get onError fallback via ImgWithFallback component, feed images loading=lazy
- ~15 emoji-prefix names render correctly via new safeInitials util

R15**8 fixes****PWA + cross-screen state coherence + audio**

- **CRITICAL:** Blocked users' posts kept appearing in feed + their moments kept playing (block button worked but had ZERO consumers)
- **3 stale-closure regressions** defeating R2 restrict — inbox channel, chat channel, App-level global badge all captured the restricted set at subscribe time
- Manifest maskable icons dropped (React-atom had no safe zone), ringin-name-changed broadcast, ringtone visibilitychange stop

R16**9 fixes****Lifecycle cleanup + network resilience + auth edges**

- **P0:** handleAuth no try/catch (offline login stuck on "Please wait..." forever), agora fetchToken no timeout (call hung on "Connecting..."), fcm_token never cleared on SIGNED_OUT (next user got prev user's pushes)
- onMessageListener leak (notification sound fired N× after token refreshes), HomeScreen notifs effect dep stable across token refresh
- 2 password .catch handlers (Reset + Change buttons no longer stuck), 2 typingTimer cleanups

R17**8 fixes****Search/discovery + animations + empty states**

- 3 superseded-search races (fast typing → old query clobbers new results) — added seqRef guards in HomeScreen + SearchScreen FTS + MessagesScreen people-search
- MessagesScreen people-search undebounced + no .catch + no empty state — fixed all 3
- HomeScreen feed empty state ("No posts yet"), country + timezone picker empty state ("No matches"), hashtag-no-match empty state
- FollowersListModal stale users on reopen, MessagesScreen new-msg toggle clears stale search

R18**11 fixes****Storage failures + modal a11y + broken-image leaks + form-abort**

- Toast dedupe (1s window per type+message) + visible cap at 3 — rapid error spam no longer stacks endlessly behind the bottom nav
- ESC closes 9 modals (ReportModal, phone-code picker, timezone picker, FollowersListModal, post-detail, edit-post, notifications panel, comments panel, chat 3-dot menu) + body-scroll-lock on each
- safeSetItem wrap on 12 unwrapped localStorage.setItem sites (ProfileScreen avatar cache + acct fields, HomeScreen moments sync + chat caches, SavedPostsScreen) — Safari private mode no longer crashes the screen
- MessagesScreen clearAllChat killed native window.confirm — replaced with in-app modal (CLAUDE.md explicitly banned native dialogs since they block JS thread + look system-y on Android)
- ReportModal: 10s AbortController timeout + 60s per-target dedupe + X button stays clickable mid-submit + reset on target change
- New shared ImgWithFallback component replacing 15+ bare tags: avatars in FollowersListModal, SavedPostsScreen, MessagesScreen (inbox + chat + new-msg picker), CallScreen (declined/ringing/connected), Moments (share-sheet/viewer/cube); chat-image bubbles get "Image unavailable" fallback
- 4 timezone-aware date sites (HomeScreen notifications + post timestamps, SavedPostsScreen, WalletScreen transactions) via new formatDate/dateTime/formatDate/formatTime helpers

- **useFollow rapid-tap race + .catch on offline** — pendingRef Set guards parallel writes; without it tap-tap-left state flipped (out-of-order resolvers) and offline left optimistic state stuck forever
- **Blocks broadcast event** — block from chat / Moments / Profile-blocked-list now dispatches ringin-blocks-changed so HomeScreen feed-filter, MessagesScreen send-guard, Moments strip all re-read immediately (was stale until app reload)
- **Muted-posts broadcast event** — mute from HomeScreen / UserProfileView / ProfileScreen post-menu all dispatch ringin-muted-posts-changed + cross-tab storage event listener so the other screen's menu label + state stay in sync
- **3rd formatTime NaN + negative-diff guard** in MessagesScreen — HomeScreen.timeAgoUtil and MessagesScreen.timeAgo already had this from R11 FIX #2, but the inbox-row formatter was missed. No more "-1m ago" on fresh PWA installs (server clock ahead of client) or "NaNm ago" on invalid timestamp
- **Moments toast timer ref + unmount cleanup** — 1.4s setSentToast(' ') was firing on dead component if user closed viewer mid-toast. Also clear-and-restart on back-to-back toasts
- **ProfileScreen acctSaved 3 setTimeout sites → ref-tracked helper** — was firing setState on dead subtree if user navigated away mid-Save
- **IncomingCallModal prefetched-token cleanup** — window.__ringinPrefetchedAgoraToken was lingering after reject. **Verifier caught:** my first attempt deleted the stash on ALL unmounts, including Accept — which defeated the ~200-500ms prefetch optimization because CallScreen reads the stash AFTER an rAF yield (post-unmount). Fixed by adding acceptedRef set inside accept() before props.onAccept triggers unmount.
- **blocks-changed event had no consumers** (verifier caught) — added _resyncFromLegacyLS in blocks.js that listens to the window event + cross-tab storage event, merges legacy LS IDs into _cache, notifies React subscribers. HomeScreen now subscribes via onBlocksChanged → blocksTick bump → feed re-filters via isBlockedSync.
- **toggleMutePost dispatch inside setMutedPosts updater** (verifier caught) — StrictMode double-invokes updaters in dev → event fired twice. Moved dispatch outside the updater; setState still uses functional form for concurrency safety.

R20

8
fixes

Session expiry edges + DOM hygiene + multi-tab cache leakage + auth refresh overhead

- **ProfileScreen cover-save bio JSON guard** — if the bio re-read fails (session expired → RLS rejects, or network drop), bail and toast instead of merging into empty object (was silently wiping every other bio JSON field: bio text, social links, location, notif_prefs, sound_prefs)
- **NEW shared bodyScrollLock util** with reference-counted acquire/release. Replaces 6 independent per-modal snapshot/restore effects (R18 introduced) that leaked overflow:hidden permanently when 2 modals stacked (e.g. post-detail → edit-post over it, close in any order). Updated: HomeScreen×2, ProfileScreen×2, FollowersListModal, ReportModal.
- **SIGNED_OUT cache wipe extended** — adds ringin_blocks_v2, ringin_blocks_migrated_v2, follows_, fcm_token_ + new resetBlocksCache() export from blocks.js resets the live in-memory _cache Set. Next user signing in on the same device gets a clean slate.
- **FCM requestNotificationPermission** — now gates on Notification.permission BEFORE calling requestPermission. Bail immediately if denied (browser won't re-prompt anyway). Default state asked once per session. Profiles.fcm_token write only fires if token differs from cached fcm_token_ in LS (no more hourly redundant DB writes on TOKEN_REFRESHED).
- **otaUpdater _otaStarted guard** — module-scope flag prevents double-registration of the setInterval + visibilitychange/online listeners when App.js's main useEffect re-runs on auth state change.
- **MessagesScreen tap-on-active-tab in-place reset** — removed key='messages-' + msgResetKey from MessagesScreen mount. The remount triggered a realtime-channel collision (new instance subscribed before old cleanup ran → cleanup of OLD removed channel NEW had adopted → chat stopped updating until next nav). Now dispatches ringin:messages-reset event → existing MessagesScreen closes active chat + scrolls inbox top in-place.
- **SW registration cache-busting** — added ?v=v20 query param + updateViaCache: 'none' registration option so deploys reliably push new SW versions even if Vercel ever changed cache headers on /sw.js. Bump per round.
- **Chat photo raw-fallback insert .catch** — was only handling r.error, leaking temp message + blob URL on raw network reject (offline). Now mirrors the .error branch cleanup in .catch.

R21

7
fixes

Deep-link push handling + lifecycle battery + cross-account cache integrity + native-dialog hygiene

- **Lock-screen push Accept hoisting fix** — `actionParam==='accept'` branch tried to call `acceptIncomingCall` declared LATER in the component (var-hoisting made it always undefined). User saw nothing on tap. Now always opens the ring modal (one extra tap, always works).
- **OTA check skipped while backgrounded** — Capacitor WebView persists across pause/resume, so the 5-min interval was hitting GitHub for hours while user wasn't even looking. `doCheck()` now early-returns if `document.hidden`. R20 FIX #6 (start-once guard) preserved.
- **Coin balance cross-account leak** — `setSharedCoinBalance` was always writing the legacy global LS key in addition to the per-user key. User A signs out → user B signs in on same device → `getCachedCoinBalance()` (called from App.js call fallback before B's hook fetches) returned A's value. B's CallScreen mounted with wrong balance, per-minute deductions wrecked B's real coins. Fix: only write legacy key when no `userId` (anon bootstrap).
- **2 agora.js alert() → toastError** — mic-unavailable + audio-publish-failure were using `alert()` which blocks the JS thread mid-WebRTC handshake (Android Chrome) and pauses iOS PWA `AudioContext` (kills next call's audio session). CLAUDE.md explicitly bans `alert()`.
- **10 ProfileScreen + 2 MessagesScreen alert() → toast** — profile image type/size validation, profile save error, restrict/unrestrict failure, account-deletion result, OTA check result. All 12 were blocking the JS thread + looked out-of-place on Android.
- **String() coerce on 4 name.split(' ') sites** (Moments composer placeholder + chat-menu restrict/block buttons) — `truthy` check passes for numbers/arrays but `.split` crashes. If a corrupted `profile.full_name` is non-string, the Moments viewer and chat menu now don't crash inside the `ErrorBoundary`.
- **App.js prevTab reset on top-level nav** — tapping the connect-orb sets `prevTab='connect'`; if user then taps Experts top-tab, back-from-Experts would land in Anonymous Connect instead of home (confusing). Now `setPrevTab('home')` fires before `setActiveTab(tab.id)` on every top-level nav tap; `openWallet/connect` overrides still work.

Patterns we fixed at the root

Scope leaks 7 sites caught

Variable declared in one component used from a sibling component scope. Multiple types: mutedConvos declared in ChatBox used from MessagesScreen, userPosts declared in UserProfileView used from saveEditPost, props.X used in components that destructured props. Same class of bug as the original "mutedConvos is not defined" crash. We fixed each one and audit agents specifically grep for this pattern now.

Stale closures in setInterval/setTimeout/listeners 10+ sites

hangup() in CallScreen captured by 6+ stale closures (poll, 35s timer, per-min tick), localCoins read via closure inside hangup (would have undone every per-minute deduction), restrictedSet captured at channel subscribe time, etc. Fix pattern: mirror state into ref via useEffect, read ref.current inside async callbacks.

Missing .catch on Supabase promises 15+ sites

.then(r => { if (r.error) rollback }) handles RLS denial but NOT raw network reject. Pattern shipped: every Supabase call also has .catch(e => { rollback; toastError }). Caught stuck spinners in handleAuth, processPayment, saveEditProfile, save*, toggleLike, doSearch, matchAnonymous, password reset/change, follow/unfollow, mark-read, delete, ...

Optimistic UI without rollback 8+ sites

setState(optimistic); supabase.write() with no rollback path on failure. Each one was a silent data corruption: setUnreadNotif(0) without restoring on error, deletePost without snap restore, etc. Pattern shipped: snapshot before, restore on error in both .then(r => r.error) AND .catch.

Stuck spinners on network failures 5+ sites

Loading state set true → async call → on failure, never set false → user must hard-reload. Audited every "Loading..." / "Sending..." / "Processing..." state. Fix: always reset in catch.

Hardcoded "1,240" coin balance 5 sites + CallScreen prop

Coin chip showed "1,240" everywhere across screens. Each screen had its own useState(50) or hardcoded literal. Built useCoinBalance shared hook with per-user storage + Supabase realtime + cross-screen broadcast. App.js feeds real balance to CallScreen so call deduction starts from actual wallet.

Cross-tab / cross-component event leaks ~10 listeners

Every 'ringin:*' custom event verified: producer + consumer + cleanup. Window events: ringin:back, ringin:back-call, ringin:restricted-changed, ringin:muted-convos-changed, ringin:coin-balance, ringin:open-post-detail, ringin:home-open-post, ringin:incoming-call, ringin:name-changed. All have matching addEventListener / removeEventListener.

Items deferred (need your manual input)

These can't be finished autonomously — require Supabase Console / Stripe account / Firebase Console / Apple Developer access:

ITEM	NEEDS	WHY DEFERRED
Workshops payment flow	Stripe or Razorpay account + secret keys	Can't take real money without merchant config
LiveWorkshopScreen real session	New Supabase schema (workshop_sessions, workshop_messages tables)	You haven't asked for these tables; spec uncertain
Firebase API key rotation	Firebase Console access	Keys currently committed in repo + CLAUDE.md
Email confirmation in auth flow	Supabase Auth settings + email template config	Requires admin Supabase access
iOS push notifications	APN .p8 key + Apple Developer (\$99/yr)	Can't be done without paid Apple account
Real EXPERTS Supabase table	Schema for expert profiles + verification flow	You haven't approved a schema
profile_vis / profile_locked / notif_* RLS	Supabase RLS policies + push routing infra	Backend-side enforcement
WorkshopsScreen JSX → React.createElement	Code style migration only	Cosmetic, working as-is
OTA Capgo undefined-resolve edge	Capgo plugin internal investigation	Rare edge, requires repro work
popstate empty-URL on Firefox	Browser-specific UX tweak	Capacitor uses Chromium WebView anyway
RINGIN_BUILD debug label	Cosmetic	Gated to ringin_debug=1 only — invisible to real users

Test plan — verify on your phone

Install `~/Desktop/ringin-v3.38.apk`, accept the green Update prompt for the latest bundle, then walk through these scenarios. Each maps to a specific fix; if any breaks, that's a regression report worth sending back.

1 Edit Post from your own profile

Used to crash (P0 scope leak from R1). Modal should open, edit, save without ReferenceError.

2 Tap a liker on a post

Used to crash (P0 props destructure leak). Should navigate to user profile.

3 View My Moment from avatar menu

Used to crash (same P0 class). Should open Moments viewer.

4 Account Settings → change country/timezone → Save

Used to wipe cover/bio/notif/sound prefs (R1+R7 bio merge fix). Verify your existing fields survive.

5 Edit Profile dialog → change name → Save

Same data-loss class (R7). Verify cover/location/prefs survive AND top bar updates immediately (R15 ringin-name-changed event).

6 Wallet → Buy coins package → confirm screen shows ₹price

Was empty grey div (R8). Verify Card Number input shows numeric keyboard (R13).

7 Make a call → press Android back

Used to leave Agora session alive + DB invite stuck ringing (R8 ringin:back-call). Verify call ends cleanly + coins persist.

8 Tap Call on expert while phone is ringing inbound

Used to stack 2 calls (R12). Should be ignored.

9 Share an image-only post (no caption)

Used to crash null.substring (R8). Should share OK.

10 Block a user → their posts should disappear from feed + their moments from strip

R15 critical fix — block button finally has consumers.

11 Restrict a user → message yourself from another account → should be silent

R2 fix + R15 stale-closure fix. Should take effect IMMEDIATELY (no app restart needed).

12 Sign out → sign in as different user → no flash of previous user's data

R7 SIGNED_OUT cleanup + R16 fcm_token clear.

13 Picking a photo for Moments >10MB or non-image file

Should reject with toast, not OOM Android (R12).

14 Long carousel — swipe to slide 3 → switch tab → come back

Slide position persists (R3 carousel memory).

15 Hardware back button

Closes innermost overlay first → tab nav → home → exit. Should never exit on first press unless already on home (R6 + many follow-ups).

16 Pull-to-refresh on Home + Messages inbox

Both should work (R3 PTR scroller fix).

17 Open a chat → scroll up to read history → receive new message

View should NOT jump to bottom (R2 scroll-up guard).

18 Type a chat message in Chinese / Japanese / Korean IME → press Enter to confirm candidate

Should NOT send the half-composed message (R14 IME guard on 9 Enter handlers).

19 Open Saved Posts → tap a saved post

Used to do nothing (R10 — dispatched event had no consumer, now 3-hop chain works).

20 Sign-in offline / on flaky network

Button should reset on error, not stick on "Please wait..." (R16 handleAuth try/catch).

21 Tap a name with emoji prefix (e.g. " 🧑 Maria")

Avatar initials should render cleanly, not show corrupt half-surrogate chars (R14 safeInitials).

22 Type fast in any search input

Results should match the LATEST query, not get clobbered by an older slow query (R17 seqRef guards).

23 Empty states everywhere

New user, empty feed → see "No posts yet" message, not blank scroll area (R17).

Notes on what's NOT regression-tested

The audit + verify agents specifically check for scope leaks, hook order, missing .catch, and regression of previously-shipped fixes — but they do NOT run the actual app. Verification is static (read the code) not dynamic (tap buttons). If any fix introduced a subtle runtime bug that compiles cleanly, only your real-device testing will catch it.

If you find any regression, send me the symptom (which screen, which tap, what happened) and I'll fix it surgically in the next iteration.